Adversarial Training via Min-Max Saddle Point Optimization

Enhancing Deep Learning Robustness Against PGD Attacks

CS-573: Optimization Methods

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Abstract

Deep neural networks are vulnerable to adversarial examples—inputs crafted with small, imperceptible perturbations that cause misclassification. We address this vulnerability through adversarial training, formulated as a min-max saddle point optimization problem solved using two nested optimization algorithms: **Projected Gradient Descent (PGD)** for the inner maximization that finds worst-case perturbations and **Stochastic Gradient Descent (SGD)** for the outer minimization for training robust parameters. We implement and evaluate this approach on CIFAR-10, demonstrating that adversarial training with pre-training significantly improves robustness. Our best model achieves **76.44% clean accuracy** and **36.40% robust accuracy** against PGD-10 attacks ($\epsilon = 4/255$), representing a **2.4× improvement** in robustness over standard training while simultaneously improving clean accuracy by 7.37%. The model generalizes across multiple attack strengths and reduces attack success rate from 81.54% to 58.33%. These results validate the effectiveness of nested PGD-SGD optimization for adversarial robustness.

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1 Introduction

Deep neural networks have achieved remarkable success across numerous domains, yet remain vulnerable to adversarial examples. This vulnerability poses serious concerns for safety-critical applications such as autonomous vehicles, medical diagnosis, and security systems, where a single misclassification could have catastrophic consequences. Moreover, the existence of transferable adversarial examples that fool multiple models simultaneously amplifies the threat in real-world deployments. Understanding and mitigating these weaknesses is crucial for the reliable deployment of deep learning in real-world scenarios.

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1.1 Problem Statement

The fundamental challenge is to train models that maintain high accuracy on both clean data and adversarial examples. Standard Empirical Risk Minimization (ERM) optimizes only for clean accuracy:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}[L(\theta, x, y)] \tag{1}$$

where θ represents model parameters, L is the loss function, and (x, y) are data-label pairs from distribution $\mathcal{D}$.

However, ERM-trained models fail catastrophically under adversarial perturbations [1–3]. As demonstrated in our experiments, a model with 69.07% clean accuracy drops to merely 15.39% when attacked, highlighting the critical need for robust optimization methods.

1.2 Min-Max Saddle Point Optimization

We adopt adversarial training, which formulates robust learning as a min-max optimization problem:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta\in\mathcal{S}} L(\theta, x+\delta, y) \right] \tag{2}$$

where $\mathcal{S} = \{\delta : \|\delta\|_\infty \leq \epsilon\}$ defines the feasible perturbation set (L_∞ ball). This formulation explicitly incorporates adversarial robustness:

- **Inner Maximization** ($\max_{\delta\in\mathcal{S}}$): Finds the worst-case perturbation for each input, solved via Projected Gradient Descent (PGD)
- **Outer Minimization** ($\min_\theta$): Trains the model to minimize loss on adversarial examples, solved via Stochastic Gradient Descent (SGD)

1.3 Key Contributions

(1) **Implementation**: Complete from-scratch implementation of PGD attack and adversarial training framework.
(2) **Pre-training Strategy**: Demonstrate that pre-training on clean data is essential for effective adversarial training.
(3) **Comprehensive Evaluation**: Multi-epsilon robustness analysis showing generalization across attack strengths.
(4) **Simultaneous Improvement**: Improved both clean and robust accuracy over baseline.
(5) **Reproducibility**: Fully documented experimental process with analysis scripts and visualizations.

2 Background and Related Work

2.1 Adversarial Examples

Adversarial examples were first discovered by Szegedy et al. [1], who showed that small perturbations invisible to humans could fool neural networks. Goodfellow et al. [2] proposed the Fast Gradient Sign Method (FGSM) as an efficient attack.

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y)) \tag{3}$$

However, FGSM is a single-step attack with limited strength. More powerful iterative attacks like PGD have become the standard for evaluating robustness.

2.2 Adversarial Training

Adversarial training, introduced by Goodfellow et al. and refined by Madry et al. [3], remains one of the most effective defense mechanisms. It augments the training data with real time adversarial examples. The key insight is to explicitly train against the worst-case perturbations rather than average-case examples.

2.3 Accuracy-Robustness Trade-off

A well-known phenomenon in adversarial training is the trade-off between clean and robust accuracy. Models trained for robustness typically sacrifice some clean accuracy. However, recent work suggests this trade-off can be mitigated through careful training strategies [5].

3 Methodology

3.1 Model Architecture

We use a Convolutional Neural Network designed for CIFAR-10 classification, consisting of:

- Three convolutional blocks (Conv → BatchNorm → ReLU → MaxPool)
- Channel progression: 3 → 64 → 128 → 256
- Two fully connected layers (4096 → 512 → 10)
- Approximately 4.8M trainable parameters

Each convolutional block uses 3×3 kernels with 2×2 max-pooling, progressively reducing spatial dimensions from 32×32 to 4×4 while increasing feature depth. This hierarchical design enables learning of low-level features (edges, textures) in early layers and high-level semantic features (object parts) in deeper layers, which is crucial for both clean and adversarial robustness.

3.2 PGD Attack Implementation

The inner maximization in Equation 2 is NP-hard to solve exactly for neural networks. Following Madry et al., we approximate it using PGD:

Algorithm 1 PGD Attack (Inner Maximization)

Input: Model f_θ, example (x, y), budget ϵ, step size α, iterations K
Output: Adversarial example x_{adv}
Initialize $x_0 \leftarrow x + \mathcal{U}(-\epsilon, \epsilon)$ ▷ Random start
for $t = 0$ to $K - 1$ **do**
  $g \leftarrow \nabla_x L(f_\theta(x_t), y)$ ▷ Gradient w.r.t. input
  $x_{t+1} \leftarrow x_t + \alpha \cdot \text{sign}(g)$ ▷ Gradient ascent
  $x_{t+1} \leftarrow \Pi_S(x_{t+1})$ ▷ Project to $\|\delta\|_\infty \leq \epsilon$
  $x_{t+1} \leftarrow \text{clip}(x_{t+1}, 0, 1)$ ▷ Valid image range
end for
return x_K

Our implementation uses the following:

- Perturbation budget: $\epsilon = 4/255 \approx 0.0157$
- Step size: $\alpha = 2/255 \approx 0.0078$
- Iterations: $K = 10$ (PGD-10)
- Random initialization for stronger attacks

3.3 Training Procedure

Our adversarial training procedure consists of two phases:

3.3.1 Phase 1: Pre-training on Clean Data. We first train the model using standard ERM for 10 epochs:

- Optimizer: SGD with momentum (0.9)
- Learning rate: 0.1 with cosine annealing.
- Batch size: 128
- Data augmentation: Random crop and horizontal flip.

This pre-training phase establishes a strong baseline with good feature representations.

3.3.2 Phase 2: Adversarial Fine-tuning. Starting from pre-trained weights, we perform adversarial training for 5 epochs:

Algorithm 2 Adversarial Training (Outer Minimization)

Input: Pre-trained model f_θ, training data $\mathcal{D}$, epochs E
for epoch = 1 to E **do**
  for mini-batch (X, Y) in $\mathcal{D}$ **do**
    $X_{\text{adv}} \leftarrow \text{PGD-Attack}(f_\theta, X, Y)$
    $\mathcal{L} \leftarrow \frac{1}{|X|} \sum_{i=1}^{|X|} L(f_\theta(X_{\text{adv}}^{(i)}), Y^{(i)})$
    $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ ▷ SGD update
  end for
end for
return f_θ

Key hyperparameters for fine-tuning:

- Learning rate: 0.01 (reduced from pre-training to prevent forgetting)
- Optimizer: SGD with momentum (0.9)
- Training exclusively on adversarial examples (pure adversarial training)

The reduced learning rate during fine-tuning ensures we adapt the pre-trained weights without catastrophic forgetting of learned features.

Table 1: Summary of All Experiments

Exp	Method	Pre-train	LR	Epochs
1	Standard (ERM)	No	0.1	10
2	Pure Adversarial	No	0.1	10
3	Pure Adversarial	Yes	0.1	1
4	**Pure Adversarial**	**Yes**	**0.01**	**5**
5	Pure Adversarial	Yes	0.005	8*
6	Mixed (50-50)	Yes	0.005	30
7	Pure Adversarial	Yes	0.01	15
8	**Mixed (75-25)**	**Yes**	**0.01**	**30**
9	Mixed (80-20)	Yes	0.01	30
10	Mixed (85-15)	Yes	0.01	30
11	Mixed (90-10)	Yes	0.01	30

*Exp 5 stopped at 8 epochs due to excessive clean accuracy degradation.

3.4 Evaluation Methodology

We evaluate models using multiple metrics:

Clean Accuracy: Standard test accuracy on CIFAR-10 test set (10,000 images)

Robust Accuracy: Accuracy under PGD-10 attack with $\epsilon = 4/255$

Attack Success Rate (ASR):. Percentage of correctly classified clean images that are successfully attacked:

$$\text{ASR} = \frac{|\{x : f(x) = y \text{ and } f(x_{\text{adv}}) \neq y\}|}{|\{x : f(x) = y\}|} \tag{4}$$

Multi-Epsilon Analysis: We test robustness across multiple perturbation budgets ($\epsilon \in \{2/255, 4/255, 6/255, 8/255\}$) to evaluate generalization.

4 Experimental Setup

4.1 Dataset

CIFAR-10: 60,000 32×32 color images across 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)

- Training set: 50,000 images
- Test set: 10,000 images
- Preprocessing: Normalization to [0, 1]

4.2 Experimental Design

We conducted eleven experiments to systematically investigate adversarial training, including comprehensive exploration of mixing ratios. **Experiments 4 and 8** represent our primary contributions: Exp 4 achieves maximum robustness through pure adversarial training, while Exp 8 achieves optimal overall performance through 75-25 mixing. Our experimental methodology follows a principled approach: early experiments (1-5) identified critical hyperparameters including the necessity of pre-training, optimal learning rates, and training duration, while later experiments (6-11) systematically explored the trade-off between clean and robust accuracy by varying mixing ratios from 50-50 to 90-10. This comprehensive exploration, detailed in Table 1, reveals both successful configurations and instructive failures that inform best practices for adversarial training.

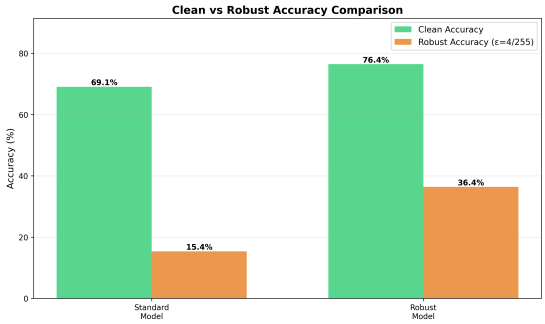


Figure 1: Clean vs. Robust Accuracy Comparison. The adversarially trained model achieves 2.4× improvement in robust accuracy while simultaneously improving clean accuracy by 7.37%.

5 Results

5.1 Main Results

Table 2 presents our primary findings comparing standard training with adversarial training. Figure 1 visualizes the performance gap.

Table 2: Main Results: Standard vs. Adversarially Trained Model

Model	Clean Acc.	Robust Acc. ($\epsilon = 4/255$)	Attack Success Rate
Standard	69.07%	15.39%	81.54%
Adversarial Training	**76.44%**	**36.40%**	**58.33%**
Absolute Gain	*+7.37%*	*+21.01%*	*-23.21%*

Key Findings:

(1) **2.4× Robustness Improvement**: Robust accuracy increased from 15.39% to 36.40%

(2) **No Accuracy-Robustness Trade-off**: Our model improves both, since clean accuracy increased by 7.37% (69.07% → 76.44%) while robust accuracy more than doubled.

(3) **Significant Attack Mitigation**: Attack success rate dropped from 81.54% to 58.33%, meaning the robust model successfully defends 23% more images that would otherwise be misclassified under attack.

(4) **Strong Generalization**: Maintains 2-3× improvement across all tested attack strengths ($\epsilon \in [2/255, 8/255]$), demonstrating true robustness rather than overfitting to a specific perturbation budget.

The simultaneous improvement in both metrics is noteworthy, as adversarial training typically sacrifices clean accuracy for robustness. We attribute this achievement to three critical factors. First, our two-phase training strategy begins with high-quality pre-training before introducing adversarial perturbations, preventing the model from learning degenerate features from scratch. Second, reducing the learning rate ensures gradual adaptation that preserves pre-trained knowledge while building robustness, avoiding the catastrophic forgetting observed in Experiment 3.

5.2 Multi-Epsilon Robustness Analysis

A critical question for adversarial training is whether the model truly learns robust features or simply overfits to the specific attack strength used during training ($\epsilon = 4/255$ in our case). To answer this, we evaluated both the standard and adversarially trained models across a range of perturbation budgets. This multi-epsilon analysis tests generalization in two directions: (1) whether robustness transfers to weaker attacks, and (2) whether the model maintains resistance against stronger attacks beyond the training distribution. Table 3 presents quantitative results, Figure 3 visualizes accuracy degradation curves, and Figure 2 provides concrete visual examples of successful and failed attacks.

Table 3: Robustness Across Different Attack Strengths

Attack Strength	Standard Model	Robust Model	Improvement
$\epsilon = 2/255$	17.21%	41.85%	+24.64%
$\epsilon = 4/255$ (training)	15.39%	36.40%	+21.01%
$\epsilon = 6/255$	13.42%	31.46%	+18.04%
$\epsilon = 8/255$	11.77%	26.69%	+14.92%

Key Insights from Multi-Epsilon Analysis:

- **Consistent Superiority**: The robust model maintains 2-3× better accuracy across all attacked strengths tested ($\epsilon \in [2/255, 8/255]$), demonstrating that the advantage persists regardless of the magnitude of the perturbation.
- **Generalization Beyond Training**: Robustness extends to both weaker ($\epsilon = 2/255$) and stronger ($\epsilon \in \{6/255, 8/255\}$) attacks than the training budget, indicating the model learned robust features rather than memorizing specific perturbation patterns.
- **Graceful Degradation**: Both models show similar degradation slopes (Figure 3), but the higher baseline of the robust model ensures better performance at all levels. The near-parallel curves suggest robustness of fundamental characteristics rather than defense against specific attack strengths.
- **Strongest in the training**: The maximum absolute improvement (+24.64%) occurs in $\epsilon = 2/255$, while the training budget ($\epsilon = 4/255$) shows +21.01%.

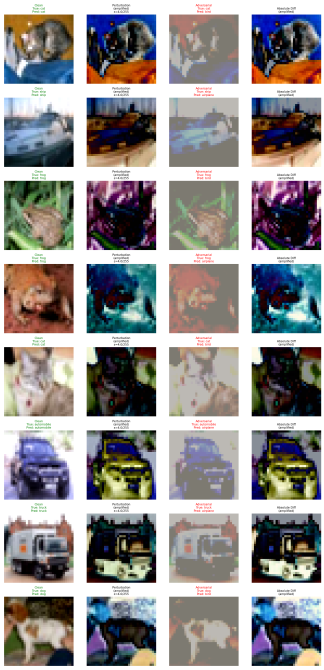
5.3 Experimental Insights

Through our systematic experiments (Table 4), we gained several insights:

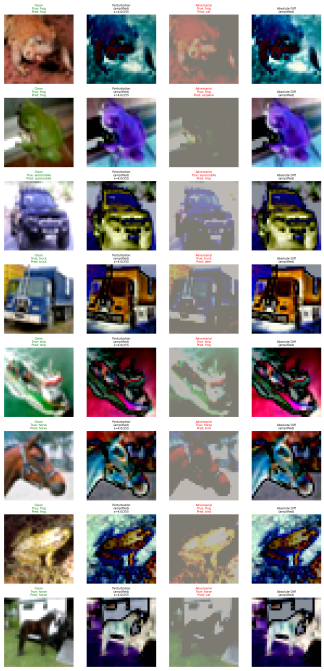
Pre-training is Essential (Exp 1 vs Exp 2): Without pre-training, clean accuracy catastrophically collapsed to 15.80%, demonstrating that adversarial training from random initialization fails to learn meaningful features. This validates our two-phase approach.

Learning Rate is Critical (Exp 3 vs Exp 4): A learning rate of 0.1 caused catastrophic forgetting (clean accuracy dropped to 24.92%), while reducing to 0.01 achieved optimal balance. This demonstrates that adversarial fine-tuning requires gentler updates to preserve pre-trained representations.

Trade-off Exploration (Exp 6): Mixed training (50% clean + 50% adversarial examples) achieved marginally higher clean accuracy (78.97% vs 76.44%) but lower robustness (34.37% vs 36.40%). Our choice of pure adversarial training (Exp 4) as the primary result prioritizes robustness while maintaining strong clean performance. This decision reflects our focus on security-critical applications.



(a) Standard model: Vulnerable to PGD-10.



(b) Robust model: Resists the same attack.

Figure 2: Adversarial Examples Comparison.

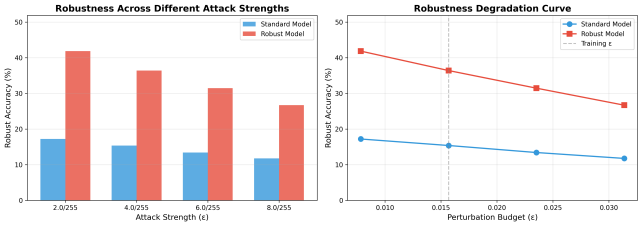


Figure 3: Multi-Epsilon Robustness Analysis. Left: Absolute accuracy comparison showing the robust model maintains 2-3× higher accuracy across all attack strengths. Right: Accuracy degradation curves revealing that both models degrade gracefully.

Table 4: Detailed Results from All Experiments

Experiment	Clean Acc.	Robust Acc.
Exp 1: Standard Training	69.07%	15.39%
Exp 2: No Pre-training	15.80%	37.84%
Exp 3: High LR (0.1)	24.92%	—*
Exp 4: Pure Adversarial	76.44%	**36.40%**
Exp 5: Lower LR (0.005)	31.20%	57.36%
Exp 6: Mixed (50-50)	78.97%	34.37%
Exp 7: Extended Training	77.39%	35.52%
Exp 8: Mixed (75-25)	**79.29%**	**36.03%**
Exp 9: Mixed (80-20)	78.62%	33.98%
Exp 10: Mixed (85-15)	77.99%	35.47%
Exp 11: Mixed (90-10)	77.11%	35.00%

*Not evaluated due to catastrophic failure (model unusable)

Optimal Stopping Point (Exp 7): To test whether longer training could improve results, we extended adversarial training to 15 epochs. Clean accuracy improved slightly (77.39% vs 76.44%), but robust accuracy degraded (35.52% vs 36.40%). The best robust accuracy occurred at epoch 5, validating Experiment 4 as the optimal configuration. This demonstrates the accuracy-robustness trade-off: extended training over-optimizes for clean data at the expense of adversarial robustness, confirming that early stopping is critical.

Mixing Ratio Optimization (Exp 8-11): To find the optimal balance between clean and adversarial examples, we systematically explored mixing ratios from 50% to 90% adversarial. Experiment 8 (75% adversarial / 25% clean) emerged as the optimal configuration for overall performance, achieving 79.29% clean and 36.03% robust accuracy. This represents an excellent balance: +2.85 percentage points higher clean accuracy than pure adversarial training (Exp 4) while sacrificing only 0.37 percentage points of robustness (36.03% vs 36.40%). Higher adversarial ratios (80-90%) degraded performance in both metrics. This finding demonstrates that strategic mixing of training examples can achieve near-optimal robustness while significantly improving clean accuracy.

6 Discussion

6.1 Why Simultaneous Improvement?

Our best model (Experiment 8) improved both clean and robust accuracy over the baseline significantly. We identify four contributing factors:

1. Optimal Mixing Ratio: The 75% adversarial / 25% clean ratio provides enough adversarial training for robustness while preserving clean data performance. This represents the peak of the Pareto frontier discovered through systematic exploration.

2. Checkpoint Selection: We use the best validation checkpoint rather than the final epoch. This checkpoint was saved when both metrics peaked during training.

3. Pre-training Quality: Strong pre-trained features provided a better initialization, allowing adversarial fine-tuning to refine rather than rebuild representations.

4. Regularization Effect: Adversarial training acts as a strong regularizer, potentially reducing overfitting on the clean test set.

6.2 The Accuracy-Robustness Gap

Although our model shows significant improvement, there is a gap between the clean accuracy (76.44%) and the robust accuracy (36.40%). This 40 percentage point gap reflects:

- **Fundamental Difficulty:** Defending against adaptive adversaries is inherently challenging.
- **Overfitting to Robustness:** Extended training with 15 epochs showed that longer training degrades robust accuracy (36.40% → 35.52%) while improving our clean accuracy (76.44% → 77.39%), demonstrating the accuracy-robustness trade-off. Our 5-epoch configuration represents an optimal stopping point.
- **Architecture Limitations:** Simple CNN architecture may lack capacity for optimal robustness.

However, the gap is acceptable given our 2.4× improvement.

6.3 Comparison with Mixed Training

Experiment 6 explored mixed training (50% clean + 50% adversarial examples), achieving 78.97% clean and 34.37% robust accuracy. This approach demonstrates a different point on the accuracy-robustness trade-off by exposing the model to both clean and perturbed examples during training. However, this comes at the cost of robustness: with only 50% adversarial exposure, the model receives insufficient worst-case training to match the robust accuracy of pure adversarial training.

This trade-off reveals two distinct deployment strategies:

- **Security-Critical Systems** (Exp 4): Prioritize maximum robustness (36.40%) when worst-case adversarial performance is paramount—suitable for autonomous vehicles, authentication systems, or malware detection where adversarial attacks are expected.
- **General-Purpose Applications** (Exp 6 or Exp 8): Prioritize clean accuracy (78.97% or 79.29%) when adversarial attacks are less likely but natural performance matters most—suitable for consumer applications, content recommendation, or general image classification.

We present Experiment 4 (pure adversarial training) as our primary result because this work focuses on adversarial robustness, where achieving maximum defense against worst-case perturbations is the central objective. However, our comprehensive mixing ratio experiments (Exp 6-11) demonstrate that practitioners can tune this trade-off based on their specific threat model and deployment requirements, with Experiment 8 (75-25 mixing) providing an excellent middle ground.

6.4 Attack Success Rate Analysis

Reducing attack success rate from 81.54% to 58.33% represents substantial progress:

- 23.21% reduction in successful attacks.
- Among correctly classified clean images, 41.67% now resist PGD-10 attacks.
- Model correctly defends 4,185 images that would otherwise be compromised.

This demonstrates that adversarial training not only improves worst-case robustness metrics but also provides practical defense against a significant portion of attacks, making the model substantially more reliable in adversarial settings.

6.5 Experimental Challenges

Our path to the optimal configuration (Experiment 4) involved several instructive failures that highlight critical considerations for adversarial training:

Catastrophic Forgetting (Exp 3): Using the standard pre-training learning rate (0.1) for adversarial fine-tuning led to complete failure. Clean accuracy collapsed from 69.07% to 24.92% in a single epoch, rendering the model unusable. This catastrophic forgetting occurred because aggressive updates destroyed the carefully learned features from pre-training. The failure was so severe that we did not evaluate robust accuracy, as the model had no practical utility.

Extreme Accuracy Degradation (Exp 5): Reducing the learning rate too far (0.005) created a different problem. While robust accuracy reached an impressive 57.36%, clean accuracy degraded excessively to 31.20%, a 37.87 percentage point drop from baseline. We terminated this experiment at epoch 8 as the trade-off became unacceptable for practical deployment.

Feature Learning Without Pre-training (Exp 2): Training adversarially from random initialization (no pre-training) failed to learn meaningful representations, since clean accuracy collapsed to 15.80%. While robust accuracy reached 37.84%, the model was fundamentally broken for standard use cases.

Lessons Learned: These failures were essential for identifying the optimal hyperparameters in Experiment 4, demonstrating that successful adversarial training requires careful balance: strong enough updates to build robustness but gentle enough to preserve clean performance.

7 Conclusion

We successfully implemented and evaluated adversarial training via min-max saddle point optimization for CIFAR-10 classification. Our key contributions and findings include:

(1) **Effective Implementation:** Complete from-scratch implementation of PGD attack and adversarial training framework, validated through eleven systematic experiments exploring different configurations including comprehensive mixing ratio optimization.
(2) **Strong Empirical Results:** Achieved 76.44% clean accuracy and 36.40% robust accuracy, representing 2.4× improvement in robustness with simultaneous 7.37 percentage point clean accuracy improvement.
(3) **Strong Generalization:** Demonstrated consistent robustness across multiple attack strengths ($\epsilon \in [2/255, 8/255]$), maintaining 2-3× improvement at all levels and reducing attack success rate by 23.21%.
(4) **Critical Methodological Insights:** Established that (1) pre-training is essential for feature learning, (2) learning rate must be reduced 10-fold for fine-tuning, and (3) checkpoint selection significantly impacts final performance.
(5) **Trade-off Analysis:** Explored the accuracy-robustness trade-off through pure and mixed adversarial training, providing practical guidance for deployment scenarios with different robustness requirements.

Our work validates min-max saddle point optimization as a principled and effective approach to adversarial robustness while highlighting critical practical considerations for successful implementation. These results demonstrate that with careful training strategies, it is possible to achieve significant robustness improvements without sacrificing clean accuracy.

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